

Exercise Sheet

Question 1

The table below lists three different estimators of the Bernoulli mean parameter, where n is the sample size, $S = \sum_{i=1}^n X_i$ is the sum of Bernoulli samples. In deriving the MAP and the Bayesian estimators, a $Beta(\alpha, \beta)$ prior distribution was used. (For derivation details see the Tutorial slides from week 2).

Method	Estimator
Maximum Likelihood	$\hat{\theta}_{ML} = S/n$
Maximum A Posteriori	$\hat{\theta}_{MAP} = \frac{S+\alpha-1}{n+\alpha+\beta-2}$
Bayesian posterior mean	$\hat{\theta}_{Bayes} = \frac{S+\alpha}{n+\alpha+\beta}$

1. Determine the bias and variance of these estimators.
2. How does the choice of prior influence the bias and variance of the MAP and Bayesian estimators? Which choices of prior parameters α, β reduce the variance of the estimators, and what happens to the bias with those choices. To answer this question, try at least 3 different values for α, β .
3. Compare the bias and variance of the Bayesian estimator to those of the ML and MAP, and comment on the potential implications of your findings for practical applications.

Question 2

- Derive the MLE and MAP estimators for the parameter (λ) of a Poisson distribution (hint: you've done the MLE for already in a previous exercise sheet). For the latter, use a Gamma distribution as the prior.
- Analyse and compare the bias and variance of the MLE and MAP estimators obtained.

Question 3

Suppose we have a data set for a medical diagnosis problem, and we want to use decision trees.

1. What could go wrong if your tree is as deep as it takes to classify all training points correctly?
2. How would you use confidence bounds to select an appropriate depth for the tree? (Which bounds would you use, why, and how? how would you decide this?)

Question 4

a) Suppose you trained a classifier on some data set and it achieves zero error on the training set.

- (i) Is the 0 test error necessarily a good thing? Why or why not?
- (ii) Can you use the Occam's razor bound to argue that your classifier is good? Why or why not?
- (iii) On an independent test set (which wasn't used for training) of size 30, you get 4 errors. Give a 90% and a 95% confidence upper bounds on the true generalisation error of your classifier. State what method you used to obtain it and justify why the method use is appropriate.

b) A machine learning model is trained to classify emails as spam or not spam. To evaluate its accuracy, we test it on a random sample of 1000 emails and find that it correctly classifies 850 of them.

- (i) Using the corollary of the test set bound, construct a 95% confidence upper bound for the true classification accuracy of the model.
- (ii) Interpret the notion of confidence interval in the context of email classification.